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SUPERVISED LEARNING

Task–The Machine wants to predict whether an individual will develop Type 2 diabetes based on certain health indicators.



Experience— Suppose we have a dataset of individuals with the following columns:

- ▶ x_1 : Fasting blood sugar level
- ▶ x_2 : Body Mass Index (*BMI*)
- ▶ y [Label]: Diabetes Status (0 for No, 1 for Yes)

Formal Presentation— We represent our dataset as a collection of pairs: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})\}$, where:

- ▶ $x^{(i)} = [x_1^{(i)}, x_2^{(i)}]$: is a vector representing the health indicators of the i^{th} individual.

- ▶ $y^{(i)}$ represents whether the i^{th} individual developed diabetes.

Performance—Once the model is trained, its performance is assessed on new, unseen data to gauge how well it predicts the onset of diabetes.



$([90, 22.5], 0)$



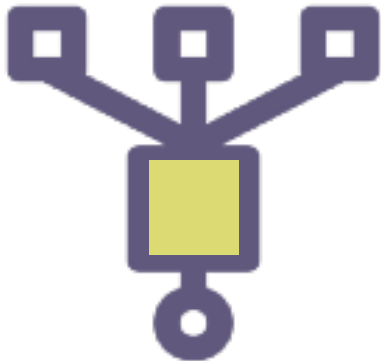
$([180, 28.7], 1)$



$([130, 25], 1)$



$([145, 27.5], ?)$



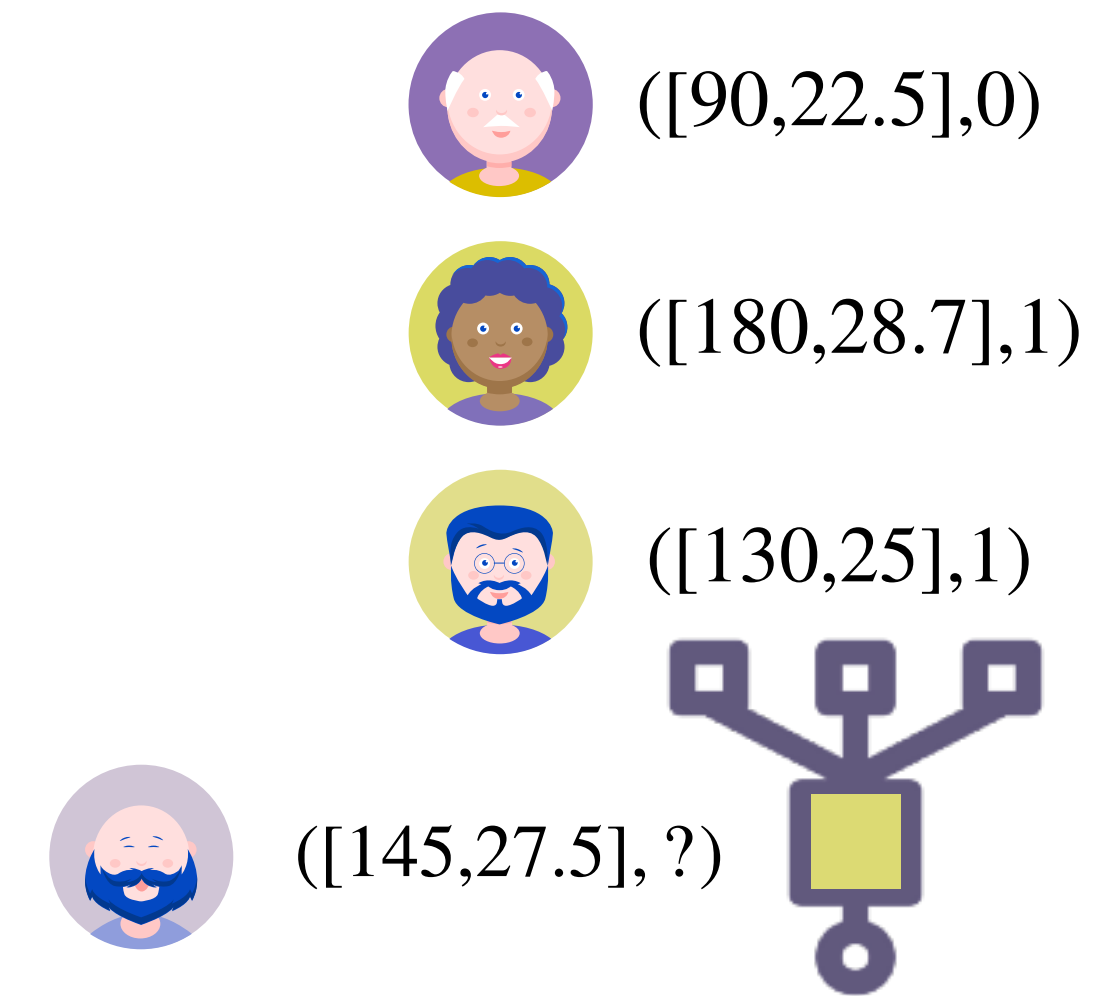
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SUPERVISED LEARNING



High-dimensional Features

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} \begin{array}{l} \text{Fasting Blood Sugar (mg/dL)} \\ \text{Body Mass Index (BMI)} \\ \text{Age (years)} \\ \vdots \\ \text{Average Daily Physical Activity} \end{array}$$

$$y = 0/1 \text{ Developed Diabetes}$$

Lecture 5? Feature Engineering: the process of selecting and transforming data attributes to enhance the performance of machine learning models.